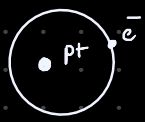
Astrofísica Básica

Interacción de la luz y la materia

El modelo atómico de Bohr

Hidrógeno



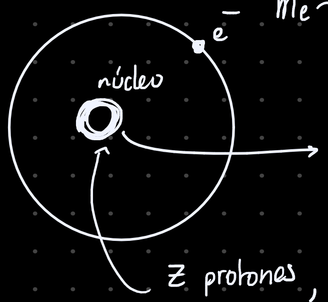
hydrogen-like

A : mass number

número de nucleones

$${}^{A}_{Z}X_{n}$$

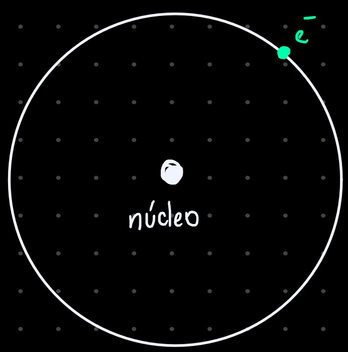
núcleo



$m_e \sim 10^{-31}$ kg $\quad -e = -1.602 \times 10^{-19}$ C

A-Z neutrones $\quad m_n \sim m_p$

Z protones, $m_p \sim 10^{-27}$ kg, $+e$



momento angular cuantizado

h constante de Planck

$h = 6.626 \times 10^{-34}$ J·s

$[J] : [N m]$

$: [kg\, m\, s^{-2}\, m]$

$: [kg\, m^2\, s^{-2}]$

$[J \cdot s] : [kg\, m^2\, s^{-1}]$

$L = mvr$

$[L] : [kg\, m\, s^{-1}\, m] = [kg\, m^2\, s^{-1}]$

fotón ν (nu)

$E = n h\nu$

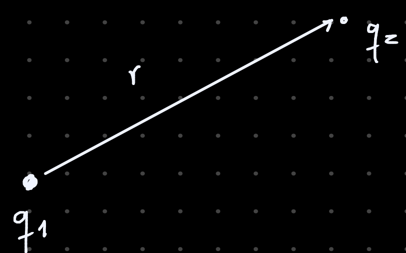
considerando $\hbar$ (hbar) $\quad \hbar = \frac{h}{2\pi}$

$L = n\hbar$

→ Ley de Coulomb $\quad q_1 \quad q_2 \quad$ distancia r

$$\vec{F} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

permitividad del espacio libre



- electrón $(-e)$
- protón $(+e)$

$$\mu = \frac{m_e m_p}{m_e + m_p} = 0.999455679\, m_e$$

$$M = m_e + m_p = 1.0005446\, m_p$$

$$\left.\begin{array}{l} \mu \approx m_e \\ M \approx m_p \end{array}\right\}$$

$$F_c = F_e$$

$$\mu \vec{a}_c = \frac{-1}{4\pi\varepsilon_0} \frac{e^2}{r^2} \hat{r}$$

$$\vec{a}_c = \frac{-v^2}{r} \hat{r}$$

$$\mu \frac{v^2}{r} = \frac{1}{4\pi\varepsilon_0} \frac{e^2}{r^2}$$

$$\mu v^2 = \frac{1}{4\pi\varepsilon_0} \frac{e^2}{r}$$

- energía cinética $K = \frac{1}{2}\mu v^2 = \frac{1}{8\pi\varepsilon_0}\frac{e^2}{r}$

- energía potencial $U = -\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r} = -2K$

→ energía total

$E = K + U$ $\quad K \geq 0$ $\quad \frac{1}{2} m v^2$ positiva positiva

$E = K - 2K$

$E = -K = \frac{-1}{8\pi\varepsilon_0}\frac{e^2}{r}$

energía cinética positiva ⟶ energía total negativa en lazados (bound)

→ cuantización del momento angular

$$L = \mu v r = n\hbar$$

$$\rightarrow K = \frac{1}{8\pi\varepsilon_0}\frac{e^2}{r} = \frac{1}{2}\mu v^2 = \frac{1}{2}\frac{(\mu v r)^2}{\mu r^2} = \frac{1}{2}\frac{(n\hbar)^2}{\mu r^2}$$

$$\frac{1}{8\pi\varepsilon_0}\frac{e^2}{r} = \frac{1}{2}\frac{n^2\hbar^2}{\mu r^2}$$

$$\frac{4\pi\varepsilon_0}{e^2} = \frac{\mu r}{n^2\hbar^2}$$

$$r_n = \frac{4\pi\varepsilon_0\hbar^2}{\mu e^2} n^2 \qquad n \in \mathbb{N}$$

$$\boxed{r_n = a_0 n^2}$$

$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{\mu e^2} = 0.0529\,\text{nm}$$

Radio de Bohr

$$\rightarrow E = -\frac{1}{8\pi\varepsilon_0}\frac{e^2}{r}$$

$$E = -\frac{1}{8\pi\varepsilon_0} e^2 \frac{\mu e^2}{4\pi\varepsilon_0\hbar^2 n^2}$$

$$E_n = -\frac{\mu e^4}{32\pi^2\varepsilon_0^2\hbar^2}\frac{1}{n^2} = -13.6\,\text{eV}\,\frac{1}{n^2}$$

número cuántico principal

$E_1 = -13.6\,\text{eV}$

$E_2 = -3.4\,\text{eV}$

$E_3 = -1.51\,\text{eV}$

↑ mundo exterior

$n=2$ first excited state $(r_2 = 4a_0,\ E_2 = -3.4\,\text{eV})$

$n=1$ $(r_1 = a_0,\ E_1 = -13.6\,\text{eV})$

núcleo ground state

n_{high} (E_{high})

γ fotón

n_{low} (E_{low})

$$\Delta E = E_{high} - E_{low}$$

$$E_\gamma = E_{high} - E_{low} \qquad \nu = \frac{c}{\lambda}$$

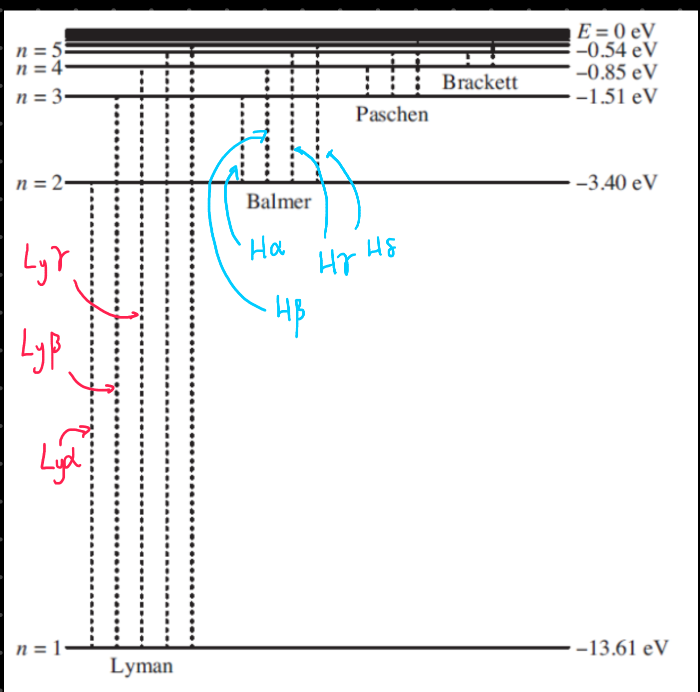
$$h\nu = \frac{\mu e^4}{64\pi^2\varepsilon_0^2\hbar^2}\left(\frac{1}{n_{low}^2} - \frac{1}{n_{high}^2}\right)$$

$$\frac{hc}{\lambda} = \frac{\mu e^4}{64\pi^2\varepsilon_0^2\hbar^2}\left(\frac{1}{n_{low}^2} - \frac{1}{n_{high}^2}\right) \qquad h = \hbar\, 2\pi$$

$$\frac{1}{\lambda} = \frac{\mu e^4}{64\pi^3\varepsilon_0^2\hbar^3 c}\left(\frac{1}{n_{low}^2} - \frac{1}{n_{high}^2}\right)$$

$R_H = \dfrac{\mu e^4}{64\pi^3\varepsilon_0^2\hbar^3 c} = 1.097\times10^{7}\ \text{m}^{-1}$ constante de Rydberg

$$\boxed{\frac{1}{\lambda} = R_H\left(\frac{1}{m^2} - \frac{1}{n^2}\right)} \qquad m < n$$



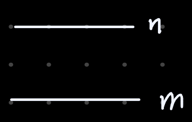
series de em/abs

			λ (nm)
Lyman	Lyα	$2 \leftrightarrow 1$	121.567
	Lyβ	$3 \leftrightarrow 1$	102.572
	Lyγ	$4 \leftrightarrow 1$	97.254
Balmer	Hα	$3 \leftrightarrow 2$	656.281
	Hβ	$4 \leftrightarrow 2$	486.134
	Hγ	$5 \leftrightarrow 2$	434.048

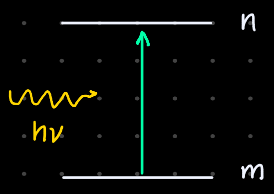
Lyman-alpha forest

Procesos atómicos

Cómo un único fotón (o electrón) interactúa con un único átomo

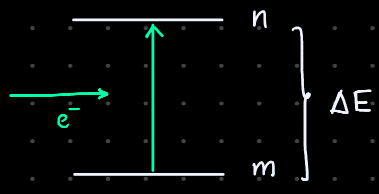


A. Absorción (fotoexcitación)



$$X + hf \longrightarrow X^*$$

B. Excitación colisional



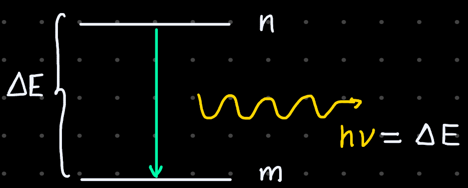
$$v > v' \geq 0$$

$$\frac{1}{2} m_e v^2 \geq \Delta E$$

$$m_e v'^2/2 = m_e v^2/2 - \Delta E$$

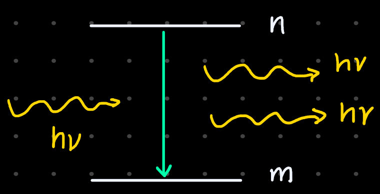
$$X + \frac{1}{2} m_e v^2 \longrightarrow X^* + \frac{1}{2} m_e (v')^2$$

C. Emisión espontánea



$$X^* \longrightarrow X + h\nu$$

D. Emisión estimulada

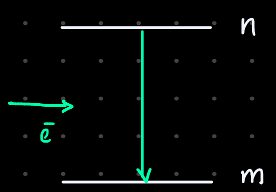


$$\nu = \frac{\Delta E}{h}$$

lasers

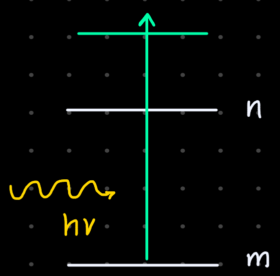
$$X^* + h\nu \longrightarrow X + h\nu + h\nu$$

E. de-excitación colisional



$$X^* + \frac{1}{2} m_e v^2 \longrightarrow X + \frac{1}{2} m (v')^2$$

F. Fotoionización

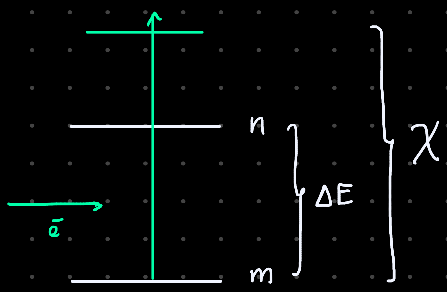


$$X + h\nu \longrightarrow X^+ + \frac{1}{2} m_e v^2$$

$h\nu > \chi$ potencial de ionización

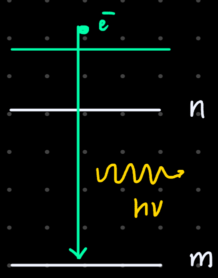
$$\frac{1}{2} m_e v^2 = h\nu - \chi$$

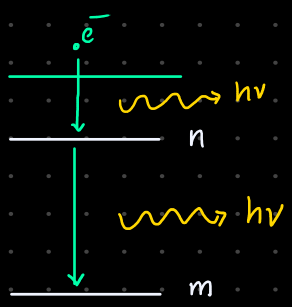
G. Ionización colisional



$$X + \frac{1}{2} m_e v^2 \longrightarrow X^+ + \frac{1}{2} m (v')^2 + \frac{1}{2} m_e (v'')^2$$

H. Recombinación





$$X^+ + \frac{1}{2} m_e v^2 \longrightarrow X + h\nu$$

$$\chi + \frac{1}{2} m_e v^2$$

Espectro de emisión y absorción